

Interaction-Aware Benchmarking of Exact-Gradient Signal and Finite-Shot Resolvability in a Four-Qubit Hybrid Quantum Neural Network

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Abstract

Purpose: We evaluate whether circuit, objective, and hybrid-connectivity interventions combine non-additively in exact-gradient signal and finite-shot resolvability for a four-qubit quantum neural network.

Methods: A complete 2^3 factorial benchmark crossed a pair-restricted CNOT schedule, normalized transverse-field Ising energy objective, and fixed residual shortcut in a measure–re-encode–reset architecture. Centered binary coding made lower-order interactions equal averages over the third factor. Configurations were matched by initialization, block count, parameter identity, and total implemented simulator-shot budget.

Results: Model-based Holm inference rejected H1–H3 but not H4. The positive centered $E \times L$ exact-gradient interaction was the strongest cross-procedure result: its initialization-cluster bootstrap excluded zero, and an independent-seed rerun retained its direction while differing in magnitude and depth profile. The centered finite-shot $E \times L$ and $E \times R$ coefficients met the Wald/Holm rule, but their achieved nested-bootstrap intervals included zero. Post-primary analyses further showed weighting sensitivity for H1 and objective- and estimator-mode sensitivity for H3.

Conclusion: Factorial benchmarking separates exact signal from finite-shot measurement behavior and exposes inference-dependent composition effects. These initialization-gradient results do not establish optimization success, trainability improvement, hardware advantage, shot savings, or matched physical-resource cost.

Keywords: quantum machine learning, quantum neural networks, factorial design, finite-shot gradient estimation, gradient resolvability, reproducible benchmarking

1 Introduction

Variational quantum algorithms and quantum neural networks (QNNs) optimize parameterized circuits through expectation values or their derivatives [4, 15, 20]. Their practical behavior depends jointly on circuit structure, objective construction, initialization, optimization, and measurement. Barren plateaus are one important failure mode, but the absence of asymptotically vanishing gradients does not guarantee that a finite-shot optimizer receives a reliable update direction [1, 13, 14, 24]. This study therefore asks a narrower operational question: whether a gradient component is resolvable under a specified finite-shot protocol.

Prior work has examined local objectives, restricted entanglement, structured initialization, hybrid architectures, and residual connections largely in separate models and against separate baselines [2, 5, 8, 10, 17, 18, 22]. Finite-shot studies likewise show that gradient precision and acquisition cost depend on the observable, circuit, and measurement allocation [3, 6, 11, 12, 23]. These results establish important marginal effects, but they do not determine whether interventions reinforce, cancel, or otherwise modify one another when combined.

A complete factorial design estimates such composition effects directly [16]. We cross three interventions that act at different points in a hybrid computation: a pair-restricted within-block CNOT schedule (E), normalized TFIM energy in place of global target-state infidelity (L), and a fixed-gain classical residual shortcut between independently reset blocks (R). The experiment holds initialization seeds, matched quantum parameters, block counts, and total shot budgets fixed across all eight configurations.

The study follows the factors, hypotheses, model families, transformations, and multiplicity procedure described in a previously published protocol article [21]. That article proposed the factorial experiment and reported no data or results from the implementation analyzed here. It did not constitute a complete preregistration: reset semantics, residual activation, plug-in-SNR eligibility, estimator-mode enforcement, and several robustness procedures were resolved during implementation or added after the primary outputs were inspected. We therefore describe H1–H4 as *protocol-derived primary analyses conditional on the reset-per-block implementation*, not as unconditional preregistered tests, and label every later validation, sensitivity, and independent-seed analysis explicitly.

The contribution is threefold. First, the experiment directly estimates non-additive effects among three QNN gradient-resolvability strategies under matched conditions. Second, it separates exact-gradient magnitude, finite-shot resolvability, estimator bias, sign reliability, task metrics, and measurement cost rather than collapsing them into a single claim of “trainability.” Third, it provides an audited and reproducible implementation with deterministic seeds, black-box gradient validation, guarded estimator-mode separation, frozen outputs, and regression tests. The primary empirical result is a positive $E \times L$ exact-gradient interaction supported by both model-based and resampling analyses. The finite-shot analogue is presented as qualified evidence because it is conditional on estimator eligibility and sensitive to inference method and block count.

2 Background and study questions

2.1 Finite-sample gradient usability

For parameter θ_k , let $\hat{g}_k(\theta)$ be the complete finite-shot gradient estimator and $\mu_k(\theta) = \mathbb{E}_{\text{shots}}[\hat{g}_k(\theta)]$ its repeated-shot mean. The operational estimator-SNR metric is

$$\text{SNR}_{\text{est}}(\theta) = \frac{|\mu_k(\theta)|}{\sqrt{\text{Var}_{\text{shots}}[\hat{g}_k(\theta) \mid \theta]}}.$$

When an exact statevector derivative is available, the complementary measure is

$$\text{SNR}_{\text{exact}}(\theta) = \frac{|g_k^{\text{exact}}(\theta)|}{\sqrt{\text{Var}_{\text{shots}}[\hat{g}_k(\theta) \mid \theta]}}, \quad \text{Bias}_k(\theta) = \mu_k(\theta) - g_k^{\text{exact}}(\theta).$$

High SNR_{est} indicates that the estimator’s own mean is distinguishable from its repeated-shot uncertainty. It does not establish unbiasedness, correct sign, convergence, or task fidelity, so these outcomes are reported separately.

The plug-in ratio is defined only when estimated repeated-shot variance is positive. The finite-shot estimand is therefore conditional on that eligibility rule. Exactly zero-variance cells are excluded without an epsilon or variance floor, their occurrence is reported as a separate outcome, and a labeled sensitivity analysis assigns zero SNR when both the estimated mean and variance are exactly zero.

2.2 Interventions and interaction hypotheses

The pair-restricted CNOT factor (E) changes the within-block schedule while holding the number of rotations and logical CNOT gates fixed. The objective factor (L) replaces global infidelity with normalized TFIM energy; it is a full objective contrast that changes spectral weighting, observable decomposition, measurement requirements, and landscape geometry, not a pure locality intervention. The residual factor (R) is a fixed-gain classical shortcut between independently reset quantum blocks and adds no quantum gates or observables.

The novelty claim is intentionally narrow: the individual interventions and metrics are established. To our knowledge, these three implemented contrasts have not previously been evaluated together in a matched factorial experiment that separates exact-gradient magnitude from finite-shot estimator behavior. The strongest prior expectation concerned $E \times L$, because the entangling schedule changes reduced-state derivatives while the objective selects and weights the derivative components entering the measured signal. The published protocol expected sub-additivity at the exact-signal level, but all tests are two-sided [21].

H1–H4 form one Holm-corrected family at family-wise $\alpha = 0.05$:

- **H1:** the $E \times L$ coefficient η_{EL} in $\text{asinh}(|g^{\text{exact}}|)$ differs from zero;
- **H2:** the $E \times L$ coefficient β_{EL} in end-to-end $\text{asinh}(\text{SNR}_{\text{est}})$ differs from zero;
- **H3:** the $E \times R$ coefficient β_{ER} differs from zero;
- **H4:** the block-count-moderated $L \times R$ coefficient β_{LRD} differs from zero.

Depth-moderation analyses beyond H4, conditional-estimator analyses, active-residual subset analyses, and configuration-level comparisons are diagnostic or exploratory.

3 Evaluation protocol and experimental setup

3.1 Reference task and objectives

The reference task is ground-state preparation for the four-site, open-boundary TFIM,

$$H = -J \sum_{i=0}^2 Z_i Z_{i+1} - h \sum_{i=0}^3 X_i, \quad J = 1, \quad h = 0.5.$$

The target ground state $|\psi_0\rangle$ and eigenvalue bounds $E_0 \approx -3.4270$ and $E_{\max} \approx 3.4270$ are obtained by exact diagonalization of the 16×16 Hamiltonian matrix. Let $\rho_0 = |\psi_0\rangle\langle\psi_0|$ and let $\rho(\theta)$ be the prepared state. The global objective is target-state infidelity,

$$C_{\text{global}}(\theta) = 1 - \text{Tr}[\rho_0 \rho(\theta)],$$

and the local objective is normalized TFIM energy,

$$C_{\text{local}}(\theta) = \frac{\text{Tr}[H \rho(\theta)] - E_0}{E_{\max} - E_0}.$$

Both objectives have the same nondegenerate target ground state, preventing target identity from confounding the comparison. The resulting factor still represents the full contrast between normalized TFIM energy and global infidelity, not measurement locality alone: the objectives also differ in spectral weighting, observable decomposition, measurement requirements, and landscape geometry. Prepared-state energy and target fidelity at initialization are therefore reported for every configuration. These are terminal-block prepared-state metrics, not trained or optimized performance.

3.2 Hybrid block architecture and entangling schedules

To avoid conflict with the objective factor L , the number of hybrid blocks is denoted by $D \in \{1, 2, 3, 4, 6\}$. Each block applies one independently parameterized R_y rotation to each qubit, followed by exactly four logical CNOT gates. D is a *hybrid block count*, not the depth of one continuously evolving quantum state: each block begins from $|0000\rangle$, is evaluated and measured independently, and passes only classically re-encoded features to later blocks. The results therefore apply specifically to this measure–re-encode–reset architecture.

Two logical CNOT schedules are used within each block:

- **Baseline** ($E = 0$): CX($0 \rightarrow 1$), CX($1 \rightarrow 2$), CX($2 \rightarrow 3$), CX($3 \rightarrow 0$), in that order. The sequential schedule contains a causal path spanning the register within the block.
- **Restricted** ($E = 1$): odd-position blocks use CX($0 \rightarrow 1$), CX($2 \rightarrow 3$), CX($1 \rightarrow 0$), CX($3 \rightarrow 2$); even-position blocks use CX($1 \rightarrow 2$), CX($3 \rightarrow 0$), CX($2 \rightarrow 1$), CX($0 \rightarrow 3$). Each block acts within two disjoint pairs, with the pairing alternating by block position.

Both schedules are applied literally, gate by gate, in the statevector simulator, with no transpiler-level fusion or reordering. Because no quantum state persists across blocks, alternating the restricted schedule changes the paired qubits at each block position but does not accumulate entanglement across D in the manner of alternating layers in one continuous circuit. Entanglement entropy and reduced-state purity are used only as finite-system diagnostics; no area-law or asymptotic scaling claim is made from four qubits.

3.3 Residual hybrid architecture

For block $\ell \in \{1, \dots, D\}$, the quantum node is

$$z_j^{(\ell)} = \text{Tr} \left[Z_j \rho_\ell \left(x^{(\ell)}, \theta^{(\ell)} \right) \right], \quad j = 0, \dots, 3.$$

The block is initialized from $|0 \dots 0\rangle$ and depends only on its own encoded input $x^{(\ell)}$ and gate angles $\theta^{(\ell)}$. For $\ell = 1, \dots, D-1$, the next input is

$$x^{(\ell+1)} = \begin{cases} W_\ell z^{(\ell)} + b_\ell + \gamma z^{(\ell-1)}, & R = 1, \\ W_\ell z^{(\ell)} + b_\ell, & R = 0, \end{cases}$$

with fixed shortcut gain $\gamma = 1$ in the residual condition and $\gamma = 0$ otherwise. W_ℓ and b_ℓ are parameterized and initialized identically across residual and non-residual conditions; the fixed shortcut adds no trainable parameters, quantum gates, or additional observables.

The boundary convention is $z^{(0)} = 0$. Therefore, the shortcut first changes a real block input at $x^{(3)} = W_2 z^{(2)} + b_2 + \gamma z^{(1)}$, which exists only when $D \geq 3$. Thus, under matched seeds, $R = 0$ and $R = 1$ are structurally identical at $D \in \{1, 2\}$ and diverge only from $D = 3$ onward.

This activation threshold was identified during the post-run implementation audit and was not specified in the published protocol article’s H3/H4 analysis [21]. It therefore does not retroactively redefine the protocol-derived target population. The original full-sweep models remain the primary analyses conditional on the reset-per-block implementation. A $D \geq 3$ active-residual analysis is reported as a sensitivity analysis, and the exact zero residual contrast at $D \in \{1, 2\}$ is reported as a structural-validity check.

3.4 Gradient assembly and validation target

The terminal block’s gradient is a direct parameter-shift derivative of the final cost. Every upstream total derivative is assembled by reverse-mode chain rule through the classical maps and quantum-node Jacobians,

$$\frac{dC}{d\theta_k^{(\ell)}} = \sum_j \frac{\partial C}{\partial z_j^{(\ell)}} \frac{\partial z_j^{(\ell)}}{\partial \theta_k^{(\ell)}}.$$

#	E	L	R	Description
1	0	0	0	Baseline
2	1	0	0	Pair-restricted CNOT schedule only
3	0	1	0	TFIM-energy objective only
4	0	0	1	Residual only
5	1	1	0	$E + L$ (H1/H2 contrast)
6	1	0	1	$E + R$ (H3 contrast)
7	0	1	1	$L + R$ (H4 contrast)
8	1	1	1	All three interventions

Table 1 Complete 2^3 factorial design

Node-level derivatives are evaluated by the two-term parameter-shift rule for Pauli-generated rotations [15, 20]; the downstream sensitivities are propagated exactly through the classical maps. Classical gradients with respect to W_ℓ and b_ℓ are analyzed separately and are not pooled into the matched quantum-gradient distribution.

The finite-difference audit treats the complete hybrid cost as a black box and compares central finite differences against the assembled total derivative. The audit design is described here; numerical outcomes are summarized in Section 4.7, with full details in the Supplementary Information.

3.5 Complete factorial design

All eight binary combinations of E , L , and R are run (Table 1).

Within each hybrid block count, all configurations share initialization seeds, matched quantum-gate parameter indices, and the same nominal shot-budget levels.

3.6 Finite-shot estimator modes

Two finite-shot modes are collected and reported separately:

- **End-to-end mode, primary for H2–H4:** forward features and node-level shifted measurements are independently re-estimated in every replicate. Nonlinear downstream re-encoding can therefore induce finite-shot estimator bias.
- **Conditional mode, diagnostic:** forward features are held at their exact statevector values while node-level shifted measurements are resampled. This isolates Jacobian-estimation noise from uncertainty caused by re-estimating and re-encoding intermediate features.

The conditional mode is not a second primary test and cannot replace an unfavorable end-to-end result. Material disagreement between the modes is interpreted as sensitivity to forward-feature sampling rather than resolved by selecting the more favorable estimator.

For every configuration, block count, initialization, matched quantum-gate parameter, and budget level $B \in \{250, 500, 1000, 2000\}$, R_{rep} independent gradient replicates are generated. Their mean, sampling variance, SNR_{est} , $\text{SNR}_{\text{exact}}$, bias, and sign agreement are computed pointwise before aggregation.

3.7 Measurement-budget accounting

The symbol B denotes the total implemented simulator-shot budget for one replicate’s complete gradient vector. The implementation floor-divides B across every required measurement job and distributes the remainder deterministically one shot at a time, so the realized total equals B in every design row. Allocations are nonnegative, deterministic, and give no production job zero shots. This equality does not match circuits, jobs, shots per circuit, physical gates, calibration, readout overhead, hardware noise, or wall-clock time.

For global target infidelity, the simulator draws directly from the exact statevector overlap probability and records one abstract overlap-observable setting. It does not construct an inverse target-state circuit or represent target-state preparation, inverse-preparation gates, or their physical cost. TFIM energy uses one shared Z-basis multinomial setting for the three ZZ terms and one shared X-basis multinomial setting for the four X terms. End-to-end mode additionally samples $D - 1$ forward-feature Z jobs and re-encodes their noisy values; conditional mode holds those features exact and intentionally excludes that variance pathway. The modes are not pooled. The implementation records shifted, forward-feature, and objective-observable jobs, settings, and realized shots per job as descriptive resource diagnostics.

3.8 Primary summaries and estimator-fidelity diagnostics

The matched quantum-gradient distribution is summarized by the median and interquartile range of pointwise SNR_{est} , the fraction of components with $\text{SNR}_{\text{est}} \geq 1$, and dependence on D and B . The threshold one is descriptive only: it indicates that the estimator mean equals or exceeds one estimated sampling standard deviation and is not treated as a necessary or sufficient condition for optimization success.

Where exact derivatives are available, the analysis additionally reports $\text{SNR}_{\text{exact}}$, signed and absolute bias, and

$$\mathbb{K}[\text{sign}(\mu_k) = \text{sign}(g_k^{\text{exact}})].$$

Exact-gradient landscape variance is calculated across matched initializations and remains distinct from repeated-shot variance at fixed parameters.

3.9 Pilot-informed replicate and initialization counts

The independent pilot begins from $R_{\text{rep}} = 30$ replicates per cell and $N_{\text{init}} = 50$ matched initializations. Replicate count is increased until bootstrap interval widths for both the signed mean and SNR_{est} meet the prespecified precision criterion and stabilize across the next increment. Bootstrap resampling follows standard nonparametric principles [7]; the criterion is motivated operationally by adaptive-shot work [23]. Because SNR_{est} is a plug-in ratio involving an estimated standard deviation, its own Monte Carlo uncertainty can remain large after the signed mean is already well resolved.

Candidate initialization counts are evaluated in increments of 25. The selected count is the smallest candidate for which the 90th percentile of the simulated 95%

confidence-interval half-width is at most 0.20 for every primary interaction coefficient. Actual convergence, failures to meet a criterion, and any recollection are Results items rather than assumed successes; the manuscript therefore uses *pilot-informed*, not *pilot-calibrated*, sample sizes.

3.10 Analysis provenance and amendments

The published protocol fixed the eight configurations, block-count and shot-budget sweeps, end-to-end estimator SNR as the primary finite-shot outcome, the mixed-effects model families, H1–H4, the asinh transformations, and Holm correction [21]. It did not fix every operational detail. Most importantly, its prose permits both a reset-per-block reading, in which only classical features cross block boundaries, and a register-persisting reading, in which the quantum state evolves continuously across blocks (Online Resource 1, Appendix A.2). We adopted the reset-per-block reading throughout this study (Section 3.2). A register-persisting implementation would be a materially different experiment, and the inaccessible full protocol text remains a provenance limitation for the original factor-coding specification.

An implementation audit found that the originally reported lower-order H1–H3 coefficients had been interpreted using direct 0/1 reference-level coding. The adopted analysis instead uses centered coding, as required for equal averaging over the third binary factor. The direct and centered forms are exact reparameterizations: fitted values, likelihood, residuals, and random effects are unchanged. What changes is the lower-order estimand when a three-way interaction is retained. Historical direct-0/1 H1–H3 values are therefore preserved only as superseded reference-level simple interactions, not as current primary factorial effects. This was an implementation–interpretation correction, not selection of a better-fitting or more favorable model. Positive-variance eligibility, estimator-mode guards, active-residual subset fits, finite-difference checks, and later robustness analyses are labeled separately.

3.11 Mixed-effects models

Let i index initialization, c configuration, p the matched quantum-gate parameter within a block count, d block count, and b shot budget. Binary factors are centered as $E_c = E - 0.5$, $L_c = L - 0.5$, and $R_c = R - 0.5$, each in $\{-1/2, +1/2\}$; depth is centered and scaled where specified. With the three-way term retained, a lower-order pairwise coefficient averages equally over the third binary factor. Thus H1 and H2 use $E_c L_c$, H3 uses $E_c R_c$ averaged across L , and H4 uses $L_c R_c$ moderated by centered depth. Historical direct-0/1 lower-order coefficients instead described simple interactions at the third factor’s reference level and are not the adopted primary estimands. The matched repeated-measures structure uses a random intercept u_i for initialization and v_{idp} for matched parameter nested within initialization and block count. The 50 initializations are the independent top-level clusters; all other rows are repeated observations.

For H1, the response is $a_{icpd} = \text{asinh}(|g_{icpd}^{\text{exact}}|)$. The model contains all intervention main effects, all pairwise products, the three-way product, centered block count, the

Analysis family	Provenance	Interpretive role
H1–H4 mixed-model Wald/Holm tests	Factors, hypotheses, transformations, and model families were inherited from the published protocol; reset semantics were resolved during implementation	Protocol-derived primary decision rule, conditional on the reset-per-block implementation
Finite-difference checks, estimator-mode guards, optimizer refits, and residual diagnostics	Implementation audit	Validation and model diagnostics; not additions to the hypothesis family
$D \geq 3$, $D \neq 1$, and include-zero eligibility analyses	Added after primary-output inspection	Post-run sensitivity analyses
Cluster-robust inference, initialization-level resampling, independent-seed rerun, and depth moderation	Initiated after the primary analysis; each plan was frozen before its corresponding result was inspected	Post-run robustness, stability, or exploratory analyses

Table 2 Analysis provenance and interpretive roles

three intervention-by-block-count terms, and the two random intercepts. The adopted coefficient is the centered $E_c \times L_c$ interaction.

For H2–H4, the response among positive-variance-eligible cells is $y_{icpdb} = \text{asinh}(\text{SNR}_{\text{est}, icpdb})$, and $s_b = \log_2(B_b)$. The protocol-derived model is

$$\begin{aligned}
y_{icpdb} = & \beta_0 + \beta_E \tilde{E}_c + \beta_L \tilde{L}_c + \beta_R \tilde{R}_c \\
& + \beta_{EL} \tilde{E}_c \tilde{L}_c + \beta_{ER} \tilde{E}_c \tilde{R}_c + \beta_{LR} \tilde{L}_c \tilde{R}_c \\
& + \beta_{ELR} \tilde{E}_c \tilde{L}_c \tilde{R}_c + \beta_D \tilde{D}_d + \beta_B s_b \\
& + \beta_{ED} \tilde{E}_c \tilde{D}_d + \beta_{LD} \tilde{L}_c \tilde{D}_d + \beta_{RD} \tilde{R}_c \tilde{D}_d \\
& + \beta_{LRD} \tilde{L}_c \tilde{R}_c \tilde{D}_d + u_i + v_{idp} + \varepsilon_{icpdb}.
\end{aligned}$$

H2 tests the centered $E_c L_c$ coefficient, H3 the centered $E_c R_c$ coefficient averaged equally over L , and H4 the centered $L_c R_c$ -depth coefficient. The $E_c L_c R_c$ term is retained. The protocol-derived model uses the full D sweep and is conditional on positive estimated replicate variance. A separate $D \geq 3$ fit reports residual-related coefficients as a post-primary sensitivity analysis.

All mixed-effects models are fit in Python 3.12.10 with statsmodels 0.14.6, REML, and an L-BFGS-first optimizer fallback order (BFGS, CG, Powell, Nelder–Mead attempted only if L-BFGS fails to converge); no custom convergence tolerances are set beyond each optimizer’s library default. A fit is flagged singular if either variance component is within 10^{-8} of zero. Every primary and sensitivity fit in this paper

converged and none was singular; an independent BFGS refit of the adopted H2–H4 model agrees with the default L-BFGS fit to 8–9 significant figures (Section 4.7). Full package versions, random-seed derivation, and bootstrap sharding are reported in the Software Availability and Reproducibility Information section below.

3.12 Inference and multiplicity

The protocol-derived primary decision rule uses model-based Wald standard errors and two-sided p -values. H1–H4 form one family, corrected by the sequential Holm procedure at family-wise $\alpha = 0.05$ [9]. Effect estimates, unadjusted confidence intervals, raw p -values, and Holm-adjusted p -values are all reported; a primary-model rejection requires the adjusted criterion. Failure to reject is not interpreted as equivalence without a prespecified smallest effect of interest and an equivalence test.

A nested matched bootstrap propagates finite- R_{rep} uncertainty into coefficient intervals [7]. Outer resampling is by complete initialization cluster, retaining all eight configurations, block counts, budgets, and parameter identities. For H2–H4, replicates are resampled within each selected initialization–configuration–parameter–block-count–budget cell, pointwise SNR is recomputed, and the model is refit. Percentile intervals use 443 completed nested-bootstrap fits. H1 uses 2,000 completed initialization-cluster bootstrap fits with zero failures. Bootstrap iterations are distinct from the 50 independent initialization clusters and are not denoted as sample size.

3.13 Post-run robustness and independent-seed rerun

Post-primary H1 analyses fit categorical depth moderation separately in the original and independent-seed datasets. Because matched parameter counts grow with depth, observation and matched-parameter weights are identical here: 0.0625, 0.125, 0.1875, 0.25, and 0.375 for $D = 1, 2, 3, 4, 6$. The primary pooled point estimate numerically matches the categorical observation-weighted point estimate; the equal-depth average answers a different question, and differing covariance structures give slightly different standard errors.

Post-primary H3 analyses derived objective-specific $E_c \times R_c$ contrasts at $L = 0$ and $L = 1$, repeated the fit at active-residual depths $D \in \{3, 4, 6\}$, tested categorical depth moderation with model-based and initialization-clustered covariance, compared end-to-end and conditional estimator modes without pooling, and performed leave-one-initialization-out diagnostics. These are explanatory or sensitivity analyses, not additions to H1–H4.

Secondary multiplicative indices are conditional on residual status:

$$J_{EL|R=0} = \frac{G_{EL}G_0}{G_E G_L}, \quad J_{EL|R=1} = \frac{G_{ELR}G_R}{G_{ER}G_{LR}}.$$

Each G follows the frozen configuration-level RMS exact-gradient construction: unique exact-gradient parameter rows are pooled before aggregation, so deeper block counts receive more parameter/observation weight. The indices are descriptive, conditional on R , and are not transformations of the centered mixed-model coefficient. Initialization-cluster percentile intervals use 2,000 completed iterations per dataset and residual

condition, with zero failures. They do not measure optimization, trainability, or shot savings.

Prepared-state task metrics comprise raw TFIM energy, normalized prepared-state energy cost, target fidelity, and target infidelity for the terminal-block state at initialization. Exact-state duplicates across finite-shot budgets are removed. Since L changes the evaluated objective rather than state preparation, paired $L = 0$ and $L = 1$ summaries are identical when all other factors are fixed. These descriptive metrics are not optimization outcomes.

3.14 Implementation audit and use of large language models

The implementation audit included black-box finite-difference validation of assembled hybrid gradients, inspection of reset semantics and residual activation, explicit estimator-mode guards, optimizer refits, deterministic seed checks, and regression testing of all principal analysis paths. Detailed audit history, superseded pooled-mode outputs, bootstrap checkpoints, and script-level provenance are reported in the Supplementary Information and repository rather than in the main narrative.

Anthropic Claude and OpenAI Codex were used during 2026 for literature-search planning, clarification of technical concepts, implementation and verification scripting, consistency checking, and feedback on author-written prose. These systems were not treated as authors or evidentiary sources. The author independently inspected the cited literature, executed and reviewed the code, verified the numerical outputs, and accepts responsibility for every claim and analysis decision.

4 Results

4.1 Calibration and finite-shot eligibility

The initialization count was 50 independent top-level clusters. Replicate-count calibration remained weakest at shallow block counts. Exactly zero estimated replicate variance occurred only for $L = 0$: 509 of 102,400 end-to-end cells and 1,324 of 102,400 conditional cells, versus none for $L = 1$. H2–H4 therefore condition on positive estimated variance; this objective-linked eligibility is retained as a limitation rather than treated as random missingness.

4.2 Centered exact-gradient interaction (H1)

The corrected centered $E \times L$ exact-gradient interaction was 0.004043 (SE 0.001081, Wald 95% CI [0.001924, 0.006162], raw $p = 0.000185$, Holm $p = 0.000739$). The 2,000-completed-fit initialization-cluster bootstrap had zero failures and a percentile 95% interval [0.000473, 0.007535]. H1 is therefore the strongest cross-procedure result.

The independent-seed estimate was 0.007726 (Wald 95% CI [0.005592, 0.009860]); its 2,000-fit, zero-failure bootstrap interval was [0.003402, 0.011923]. The positive direction was retained, but the larger point estimate and depth differences leave magnitude stability uncertain.

Post-primary depth contrasts were, for $D = 1, 2, 3, 4, 6$, respectively, -0.005251 , -0.003579 , 0.012588 , 0.004664 , and 0.003446 in the original data, and 0.001862 ,

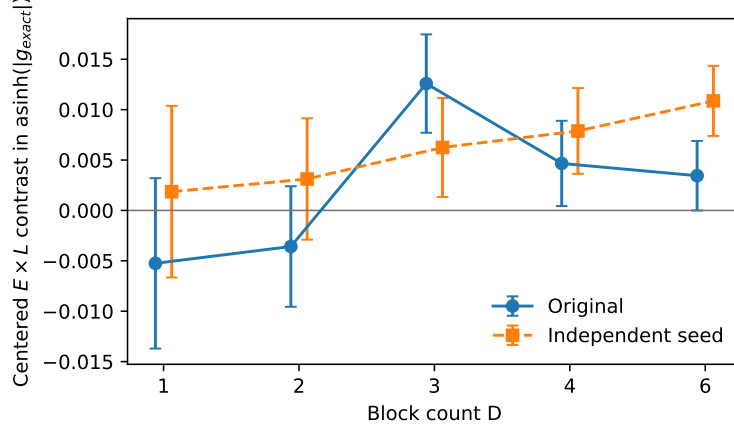


Fig. 1 Post-primary centered $E \times L$ exact-gradient contrasts by block count in the original and independent-seed datasets. Points show covariance-aware mixed-model contrasts and bars show unadjusted Wald 95% intervals. Shape and line style distinguish datasets; the horizontal reference is zero. The display does not posit a monotonic depth law.

0.003120, 0.006250, 0.007885, and 0.010870 in the independent-seed data. The original equal-depth average was 0.002374 (CI $[-0.000164, 0.004911]$), whereas its observation-weighted average was 0.004043 (CI $[0.001929, 0.006157]$). The corresponding independent-seed summaries were 0.005997 (CI $[0.003443, 0.008552]$) and 0.007726 (CI $[0.005597, 0.009854]$). Thus the original equal-depth interval includes zero but the observation-weighted interval does not; both independent-seed intervals exclude zero. Observation and parameter weights are identical. Original depth moderation was detected by the mixed model ($\chi^2(4) = 22.844$, $p = 0.000136$) but not by initialization-clustered inference ($\chi^2(4) = 6.008$, $p = 0.1985$). Neither independent-seed test rejected (mixed $p = 0.1090$; clustered $p = 0.7706$). The depth profile is consequently covariance-sensitive, and the cross-seed magnitude difference has no clear depth localization under the frozen rule.

4.3 Finite-shot interaction (H2)

Among positive-variance-eligible end-to-end cells, the corrected centered $E \times L$ coefficient was 0.014338 (SE 0.005145, Wald 95% CI $[0.004255, 0.024422]$, raw $p = 0.005321$, Holm $p = 0.015963$). The model-based Wald/Holm rule rejects H2. Its 443-completed-fit nested-bootstrap interval, $[-0.016638, 0.043563]$, includes zero. H2 is therefore procedure-sensitive and is not independently corroborated by the nested bootstrap.

4.4 Residual-related interactions (H3 and H4)

The corrected centered end-to-end $E \times R$ coefficient averaged across L was -0.011615 (SE 0.005144, Wald 95% CI $[-0.021697, -0.001534]$, raw $p = 0.023937$, Holm $p = 0.047875$). The model-based rule rejects H3, but its 443-fit percentile interval $[-0.030943, 0.010259]$ includes zero.

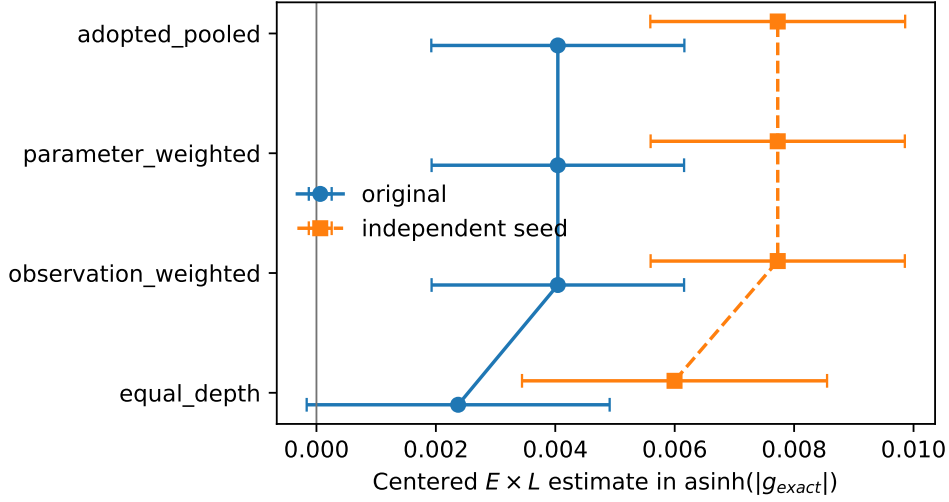


Fig. 2 Post-primary H1 weighting summaries. Equal-depth and observation/matched-parameter weighting answer different questions; the adopted pooled estimate is shown separately. Bars are Wald 95% intervals, and zero is the reference. Statistical significance is not a criterion for choosing a weighting.

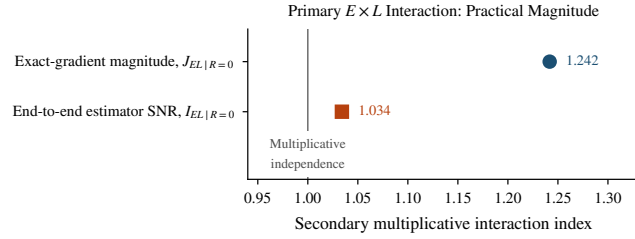


Fig. 3 Descriptive multiplicative $E \times L$ indices conditional on $R = 0$. $I_{EL|R=0}$ summarizes end-to-end estimator SNR and $J_{EL|R=0}$ exact-gradient magnitude; neither is a centered mixed-model coefficient. The line at one is the multiplicative no-interaction reference.

Post-primary end-to-end contrasts explain the averaging: at $L = 0$ the estimate was -0.000958 (CI $[-0.015251, 0.013336]$, $p = 0.896$), whereas at $L = 1$ it was -0.022273 (CI $[-0.036493, -0.008052]$, $p = 0.00214$). All corresponding 443-fit bootstrap intervals include zero. At active depths $D \in \{3, 4, 6\}$, the L-averaged estimate was -0.014788 (CI $[-0.024480, -0.005095]$); $L = 0$ remained near zero and $L = 1$ was -0.027449 . Depth-specific averages were 0.006642 , 0.000696 , -0.026492 , -0.011553 , and -0.011255 at $D = 1, 2, 3, 4, 6$. Neither model-based ($p = 0.545$) nor cluster-robust ($p = 0.721$) moderation rejected. Conditional mode reversed direction (0.006978 , CI $[-0.004745, 0.018701]$). Thus H3 is a model-dependent directional signal, objective-specific under end-to-end mode, not a robust or confirmed residual interaction.

Hyp.	Estimand	Estimate	SE	Wald 95% CI	Raw p	Holm p	Decision	Fits	Bootstr
H1	$E_c L_c$, exact	0.004043	0.001081	[0.001924, 0.006162]	0.000185	0.000739	Reject	2,000	[0.00047, 0.00616]
H2	$E_c L_c$, end-to-end	0.014338	0.005145	[0.004255, 0.024422]	0.005321	0.015963	Reject	443	[-0.0166, 0.0473]
H3	$E_c R_c$, end-to-end	-0.011615	0.005144	[-0.021697, -0.001534]	0.023937	0.047875	Reject	443	[-0.0309, 0.0077]
H4	$L_c R_c \times \text{depth}$	-0.010179	0.005362	[-0.020688, 0.000331]	0.057658	0.057658	Do not reject	443	[-0.0269, 0.0064]

Table 3 Corrected centered H1–H4 family. Decisions follow the primary model-based Wald/Holm rule. Percentile bootstrap intervals corroborate H1; H2 and H3 intervals include zero despite model-based rejection; H4 is unresolved. Fits are completed bootstrap fits, distinct from the 50 independent initialization clusters.

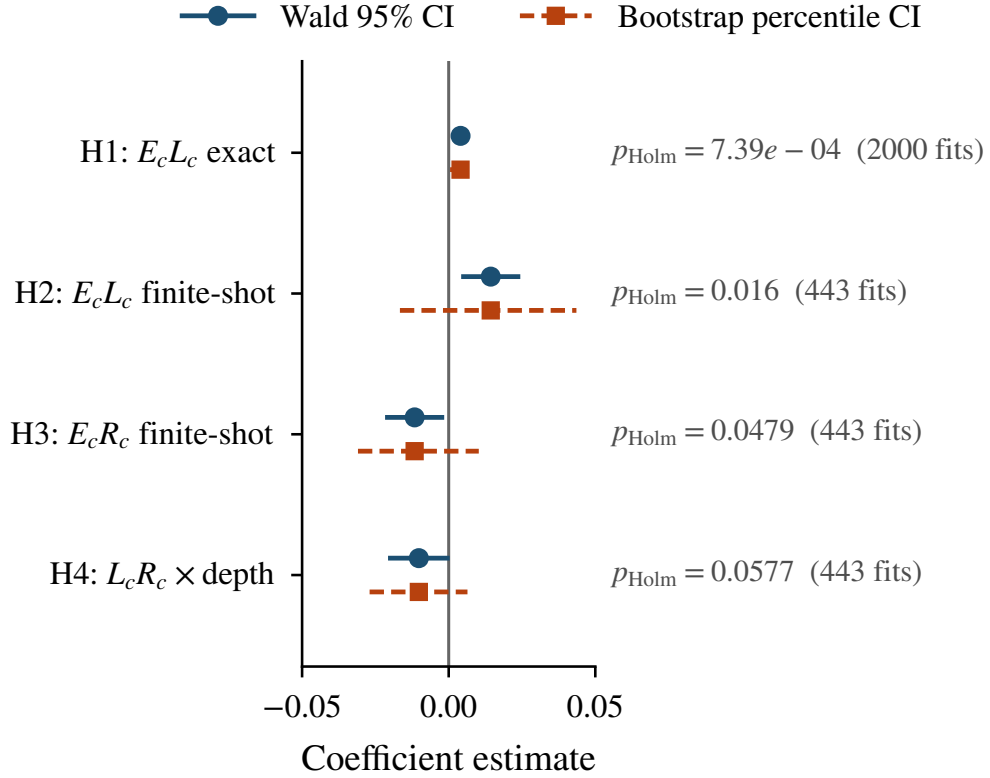


Fig. 4 Corrected primary Wald 95% intervals and initialization-cluster percentile bootstrap intervals for H1–H4, with a zero reference. H1 used 2,000 completed bootstrap fits; H2–H4 used 443. Marker shape and line style distinguish interval types without reliance on color.

The corrected centered $L \times R \times \text{depth}$ coefficient was -0.010179 (SE 0.005362, Wald 95% CI $[-0.020688, 0.000331]$, raw and Holm $p = 0.057658$). Its 443-fit percentile interval was $[-0.026975, 0.006403]$. H4 is not rejected and remains unresolved; this is not evidence of absence.

4.5 Conditional multiplicative indices

For the original data, $J_{EL|R=0} = 1.241760$ (bootstrap median 1.238674, 95% percentile CI [1.082210, 1.414864]), 24.2% above the no-interaction reference of one. At $R = 1$, $J_{EL|R=1} = 1.126633$ (median 1.124329, CI [0.991331, 1.285152]), 12.7% above one; this interval includes one. Independent-seed values were 1.163219 at $R = 0$ (median 1.160949, CI [1.007100, 1.352596]) and 1.242137 at $R = 1$ (median 1.238157, CI [1.049874, 1.449258]). Each bootstrap completed 2,000 iterations with zero failures. These weighting-dependent ratios are descriptive and conditional on R .

4.6 Prepared-state metrics and implemented resources

All bounds, uniqueness, and fidelity–infidelity complementarity checks passed for 4,000 unique terminal-block prepared states at initialization. Cell mean fidelity ranged approximately 0.0338–0.0817 in the original data and 0.0322–0.0962 in the independent-seed data; normalized prepared-state energy means ranged 0.4631–0.5529 and 0.4561–0.5537, respectively. The ranges overlap strongly and show no conspicuous seed-unstable separation descriptively. They do not demonstrate training success or a causal relationship between gradient magnitude and task quality.

All 320 resource-design rows conserve nominal implemented B , with no negative allocations or zero-shot production jobs. All 1,833 zero-variance cells joined to complete resource records. At $D = 6$, the extremes are 48 jobs for conditional/global and 61 for end-to-end/local, a 27.08% increase. This is a simulator job-count comparison, not a matched physical-resource or wall-clock claim.

4.7 Implementation checks

Black-box finite differences validated gradient assembly, centered and direct models were verified as exact reparameterizations, estimator modes were guarded against pooling, and frozen outputs passed deterministic regeneration and manifest-integrity tests. Detailed optimizer warnings, bootstrap failures, sensitivity results, and historical corrections are reported in Online Resource 1.

5 Discussion

5.1 Principal findings

H1 provides the strongest cross-procedure evidence. Its positive centered exact-gradient direction appears in both seed sets and both initialization-cluster bootstrap intervals exclude zero. Its magnitude and depth profile nevertheless remain uncertain: the seed-set point estimates differ, the original equal-depth interval includes zero while its observation-weighted interval does not, and the original moderation conclusion depends on covariance estimator.

H2 and H3 require a different interpretation. Both meet the primary model-based Wald/Holm rejection rule, yet their nested-bootstrap intervals include zero. H2 is additionally conditional on objective-linked positive-variance eligibility. H3’s end-to-end direction is concentrated in the energy-objective condition, reverses under

conditional mode, and is not supported by cluster-robust depth moderation. H4 is not rejected and remains unresolved rather than proved absent. Post-primary analyses therefore indicate that H1’s pooled positive direction is reproducible across seed sets, but its magnitude and depth profile are uncertain; H2 and H3 remain procedure-sensitive because their Wald/Holm rejections are not corroborated by the corresponding nested-bootstrap intervals.

Conditional J ratios provide descriptive multiplicative summaries, with positive above-reference patterns in several dataset–residual combinations. They are weighting-dependent, conditional on R , and not algebraic equivalents of centered mixed-model coefficients. Prepared-state metrics likewise describe initialization states only; they neither demonstrate training success nor establish that larger initial gradients causally improve energy or fidelity. Resource accounting applies to the implemented simulator protocol: total B is conserved, but job counts and shots per job differ, and no physical-resource or hardware-cost equality follows.

5.2 Interpretation and relation to prior work

Local objectives can improve gradient scaling under specified circuit-depth assumptions, while excessive random entanglement can suppress gradients [5, 17, 18]. The present result connects these observations statistically: in this four-qubit reset-based architecture, the modeled objective contrast is larger under the restricted CNOT schedule than expected from the two marginal effects on the transformed response scale. This non-additivity is compatible with theoretical accounts that treat architecture, state structure, and observable support as coupled determinants of loss concentration [19], but it neither identifies a physical mechanism nor tests asymptotic barren-plateau scaling.

The L factor also changes more than measurement locality. Normalized TFIM energy and global infidelity differ in spectral weighting, observable decomposition, circuit count, and landscape geometry. Likewise, the E factor is one specific four-CNOT schedule and the R factor is one fixed-gain shortcut between reset blocks. The observed coefficients should therefore be interpreted as interactions among these implemented contrasts, not universal laws about low entanglement, local costs, or residual quantum networks.

The results concern matched initial parameter points rather than complete optimization trajectories. A resolvable initial gradient is neither necessary nor sufficient for convergence, and residual architectures evaluated over full training trajectories address a different estimand [1, 8, 10, 22]. The unstable residual-interaction evidence here does not contradict optimization results in other architectures.

5.3 Implications for QML evaluation

The study shows why gradient-resolvability interventions should not be evaluated only one at a time. A factorial design can reveal pairwise non-additivity and moderation without implying that a fully combined configuration is universally preferable. Future benchmarks should test both whether interventions interact and whether those interactions vary with architecture-size variables.

Finite-shot metrics also require operational definitions that are treated as part of the experiment. Matching total shots does not match per-circuit precision; re-estimating intermediate features changes bias and sign reliability; and plug-in ratios can generate factor-dependent eligibility when estimated variance is zero. Eligibility rules, estimator modes, seed derivation, and regression guards should be specified and tested as carefully as the statistical formula itself.

5.4 Limitations and priorities for follow-up

The experiment covers four qubits, one open-boundary TFIM task, one reset-per-block architecture, one implementation of each intervention, one initialization family, and 50 top-level initialization clusters per dataset where applicable. Exact-state and abstract shot simulation omit hardware noise, readout and calibration models, and wall-clock behavior. There are no optimization trajectories; task metrics are terminal-block prepared-state quantities at initialization only.

Finite-replicate zero-variance selection is tied to L . Increasing parameter count with depth induces observation/parameter weighting, and some depth conclusions are covariance-sensitive. H2 and H3 disagree across inference procedures, while H3 also disagrees across estimator modes. Conditional J quantities are weighting-dependent descriptive summaries. The reset-per-block reading and inaccessible full protocol text remain provenance limitations for the original architecture and factor-coding specification. No matched physical-resource claim is available because circuits, jobs, shots per circuit, gates, calibration, readout, noise, and time were not jointly matched.

Follow-up work would require prospectively specified additional datasets, continuous-state architectural controls, optimization trajectories, and hardware-noise experiments; none is inferred from the present initialization analysis.

6 Conclusion

The strongest cross-procedure evidence concerns H1. The positive centered $E \times L$ exact-gradient interaction was retained in an independent-seed rerun, although its magnitude, depth profile, and weighting dependence remain uncertain. H2 and H3 satisfy the model-based Wald/Holm rule but have bootstrap intervals spanning zero, while H4 remains unresolved. These results support factorial and measurement-aware benchmarking of gradient usability, but do not establish optimization, hardware, or resource advantage.

Statements and Declarations

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Ethics approval. Not applicable. This study used computational simulations and did not involve human participants, human data, animals, or biological materials.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Author contributions. Brandon Shen conceived the study, implemented the simulation and analysis pipeline, conducted the experiments and verification analyses, interpreted the results, prepared the figures, and wrote and revised the manuscript.

Data availability. The public repository contains the frozen production data, pointwise gradient and SNR statistics, primary-model outputs, sensitivity analyses, independent-seed rerun, checksums, and machine-readable diagnostics used for every reported number and figure. The finalized manuscript commit is recorded in `MANUSCRIPT_COMMIT.txt`. The repository is available at <https://github.com/Brandon-Shen/few-qubit-qnn-snr>.

Code availability. The public repository contains the quantum simulation, gradient assembly, mixed-model and multiplicity code, bootstrap implementations, robustness and depth-moderation analyses, figure scripts, and regression-test suite: <https://github.com/Brandon-Shen/few-qubit-qnn-snr>.

Materials availability. Not applicable.

Software and reproducibility information. Analyses used Python 3.12.10, PennyLane 0.45.1, pandas 3.0.3, NumPy 2.5.1, statsmodels 0.14.6, SciPy 1.18.0, Matplotlib 3.11.0, PyArrow 25.0.0, and PyYAML 6.0.3. Random seeds are derived deterministically from a root seed, named stream, and cell identifiers via SHA-256. Model convergence, optimizer agreement, warnings, frozen inputs, and script-level provenance are documented in Online Resource 1 and the repository; no blanket convergence claim is inferred beyond those frozen records.

Supplementary information. Online Resource 1 contains additional estimator-eligibility analyses, model and influence diagnostics, the documented estimator-mode correction, bootstrap and depth-sensitivity results, and a reproducibility index. The supplementary file is provided as `ESM_1.pdf`.

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